

AI-GENERATED STEGANOGRAPHIC MALWARE: EVALUATING DETECTION AND EVASION IN MODERN ANTI-VIRUS SYSTEMS



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RESEARCH PROBLEM

LSB steganography conceals content across visually insignificant pixel positions. Consequently, known threat indicators may remain absent as contiguous signatures in raw PNG bytes, only becoming detectable after payload reconstruction. This study examines this critical representation gap within standard antivirus workflows.

AIM AND OBJECTIVES

This study proposes a hybrid framework to evaluate and enhance the detection of AI-generated LSB steganographic payloads. The methodology integrates VirusTotal benchmarking of raw carriers, custom YARA rule scanning, and anomaly detection utilizing an Isolation Forest trained exclusively on benign data. Detector decisions are then synthesized via logical OR/AND fusion, and the framework's overall efficiency is validated through comparative analysis on standard test images.

STUDY PROTOCOL

104 benign originals	71 / 13 / 20 train / validation / test
20 + 20 paired final test images	0 parent/test leakage

One stego derivative per held-out test original. Baseline: RGB-channel LSB, 0.30 fraction, spread placement and seed 42.

SAFETY AND SCOPE

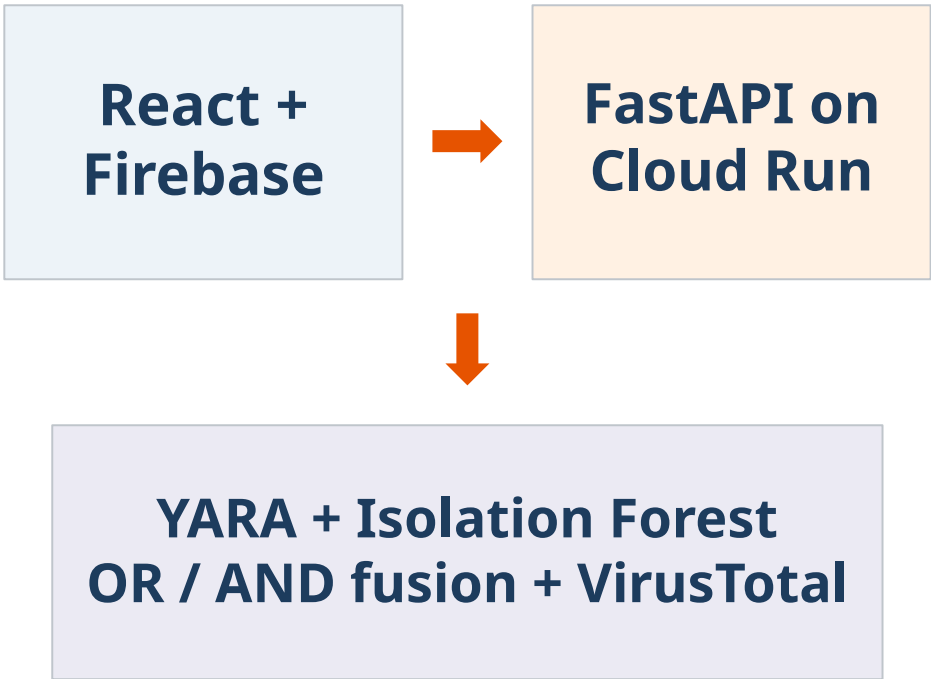
To ensure secure evaluation without payload execution, the artefact utilizes controlled AI-generated synthetic bytes and the non-executable 68-byte EICAR sequence. This EICAR sequence establishes the reported baseline to safely validate core detection mechanics.

STUDY LIMITATIONS

The evaluation used a limited image dataset and controlled LSB embedding; performance on other images, payload types and real-world conditions requires further testing.

IMPLEMENTED SYSTEM

React interface, Firebase Authentication, Cloud Firestore and Storage, Python FastAPI, and Google Cloud Run.



DETECTION METHODS

YARA

Raw carrier bytes, reconstructed payloads, or an automatic mode that selects the appropriate representation.

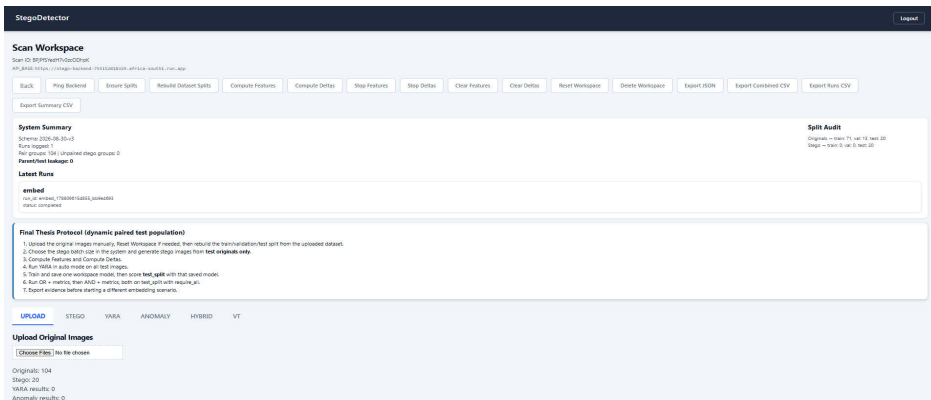
Isolation Forest

Benign-only training uses byte, colour, LSB, tile, residual and bit-plane run-length features. Benign validation data calibrate the threshold.

Fusion

OR flags either detector's positive decision. AND requires agreement.

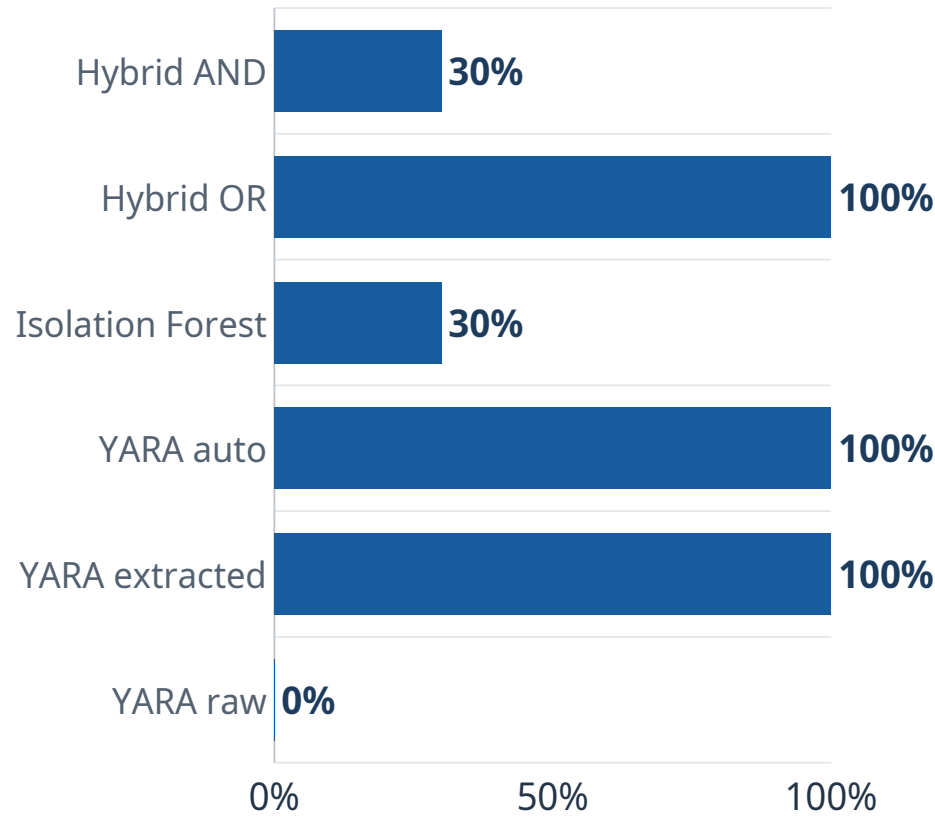
RESEARCH ARTEFACT



Persistent workspaces record image lineage, embeddings, features, models and detector outputs.

PRELIMINARY FINDINGS

Stego images lagged (%) Earlier exploratory run, n = 20



WHAT THE RESULTS MEAN

YARA detected 0/20 raw PNG carriers and 20/20 reconstructed payloads. Representation determined whether the known indicator was visible.

Isolation Forest independently flagged 6/20 stego images. Hybrid OR reached 20/20, matching payload-aware YARA without demonstrating superiority.

VirusTotal: 0 / 75

Six selected raw PNG carriers produced 450 engine decisions with no malicious or suspicious flag. This subset does not establish universal antivirus failure.

CONCLUSION AND NEXT WORK

LSB placement can hide a recognisable sequence from raw-file signature inspection. Deterministic reconstruction restores detectability. Hybrid fusion adds complementary evidence and flexible decision policies.

Further evaluation covers the paired 40-image test set, full confusion matrices, 0.10/0.30/0.50 payload fractions, placement strategies, additional synthetic payloads and image transformations.

[1] Gao et al., EURASIP JWCN, 2022. [2] Chaganti et al., Stegomalware survey, 2021. [3] Rahman et al., IEEE Access, 2022. [4] Alvarez, YARA Documentation, 2015. [5] Xu et al., IEEE TKDE, 2023. [6] Leka et al., IISA, 2022.